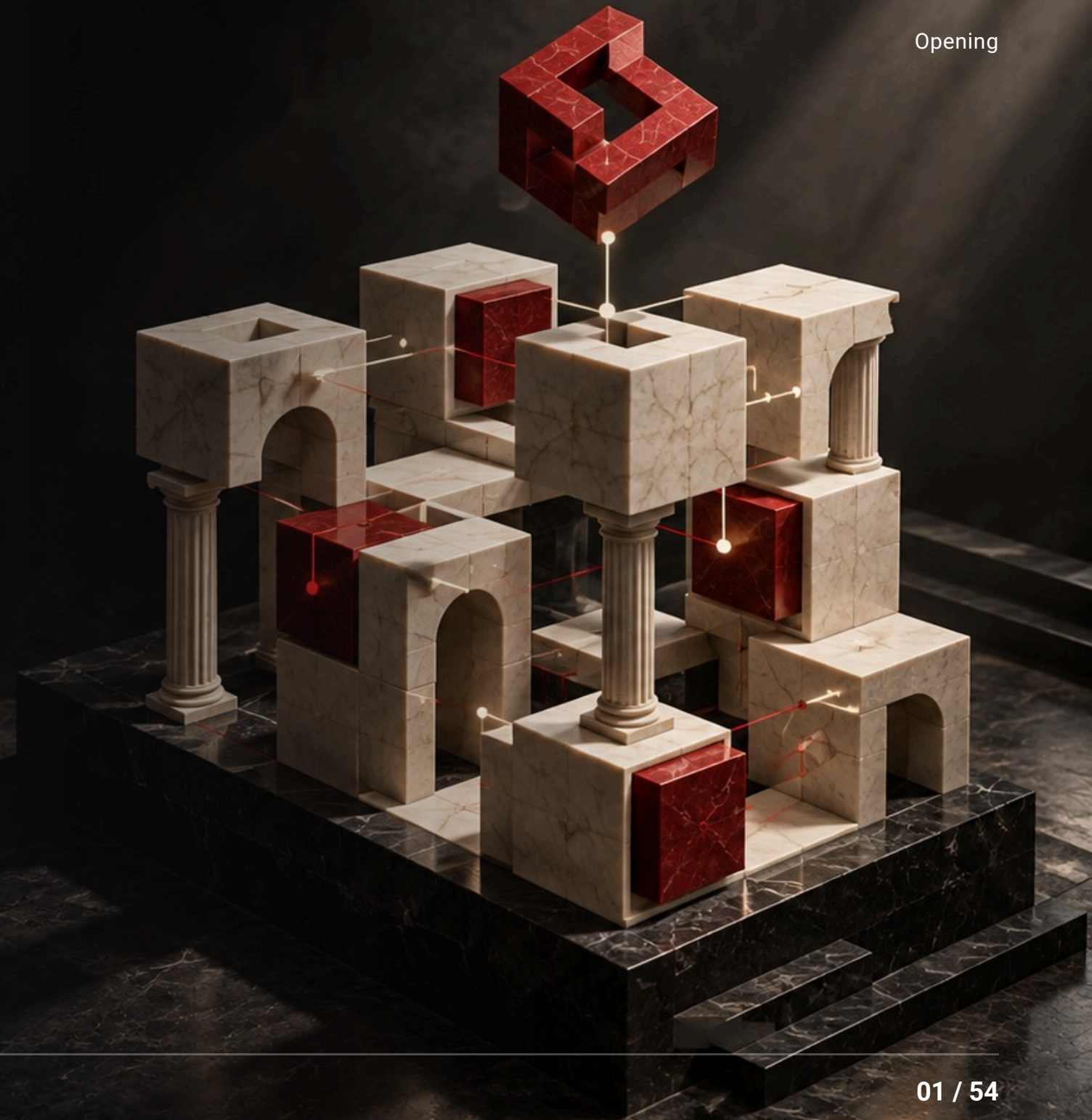


Constraint satisfaction

Unit 4 · Intelligent Systems

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Make structure do the work

54 slides · Five laboratories · Worked activities

01	02	03	04
Model	→ Search	→ Propagate	→ Repair
Variables, candidate values and relationships.	Choose a variable, try a value, recover from failure.	Remove values that have lost their support.	Start complete and reduce conflicts.

By the end, explain both a solver’s answer and the conditions that justify it.

A different representation of the same challenge

State-space search

Represent configurations, transitions and a goal test.

state → action → state

Constraint satisfaction

Represent decisions and the combinations that are allowed.

variables · domains · constraints

A CSP solution is an assignment. The order used to construct it is usually irrelevant.

The model is a triple: $\langle X, D, C \rangle$

Variables X

The decisions to make: $X_1, X_2, \dots, X_n$.

Domains D

For each X_i , a finite set D_i of candidate values.

Constraints C

Relations specifying allowed combinations of values.

A solution assigns one value to every variable and satisfies every constraint.

A domain is a set of possibilities

Initial domains

For a session S , $D(S) = \{09:00, 10:00, 11:00\}$.

An unavailable lecturer can rule out 09:00 immediately.

Current domains

After other decisions, more values may be removed.

Branch-specific removals must be restored when the branch is abandoned.

Encoding a unary restriction by reducing a domain is valid modelling.

Three useful kinds of constraint

Kind	Example	Scope
Unary	$S \neq 09:00$	One variable.
Binary	$WA \neq NT$	A pair of variables.
Global	$\text{AllDifferent}(X_1, \dots, X_k)$	A named relation over several variables.

The relation defines correctness; its propagator determines what can be pruned efficiently.

Complete and consistent are different properties

Assignment	Complete?	Violates an assigned edge?
WA=R	No	No.
WA=R, NT=R	No	Yes: WA–NT.
All seven regions set, with WA=NT	Yes	Yes.
All seven regions set, all edges unequal	Yes	No: it is a solution.

A locally consistent partial assignment may still have no valid completion.

Australia as a finite CSP

Variables

WA, NT, Q, SA, NSW, V, T.

Domains

{R, G, B} for every region.

Constraints

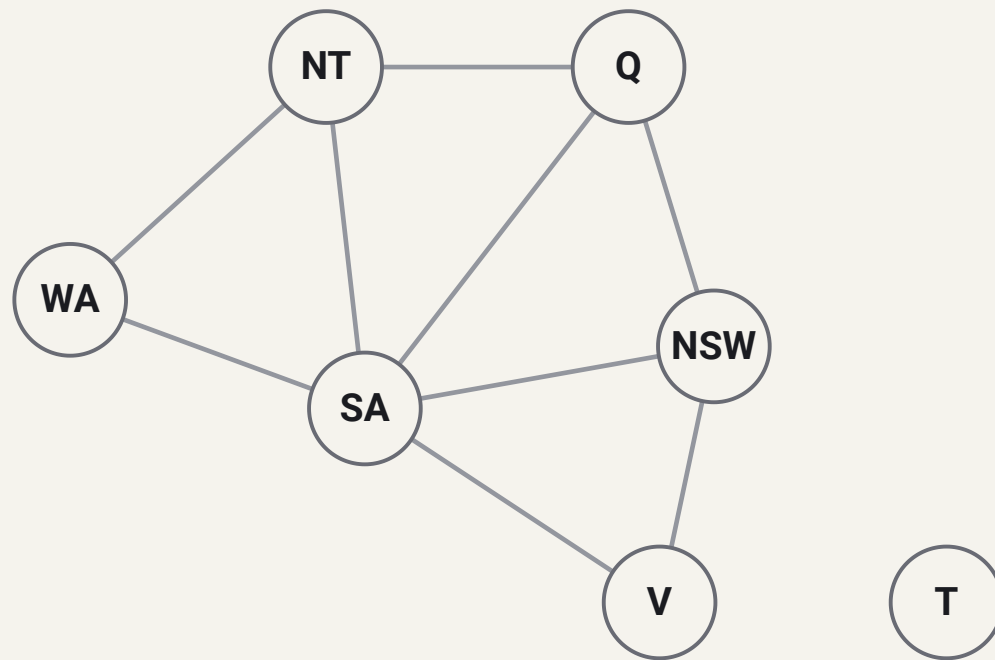
Regions sharing a land border must have different colours.

Goal

A complete assignment satisfying all nine adjacency constraints.

R, G and B are value labels. Geometric distance between nodes has no meaning.

The graph exposes the dependencies



One edge, one inequality

WA-NT · WA-SA

NT-SA · NT-Q

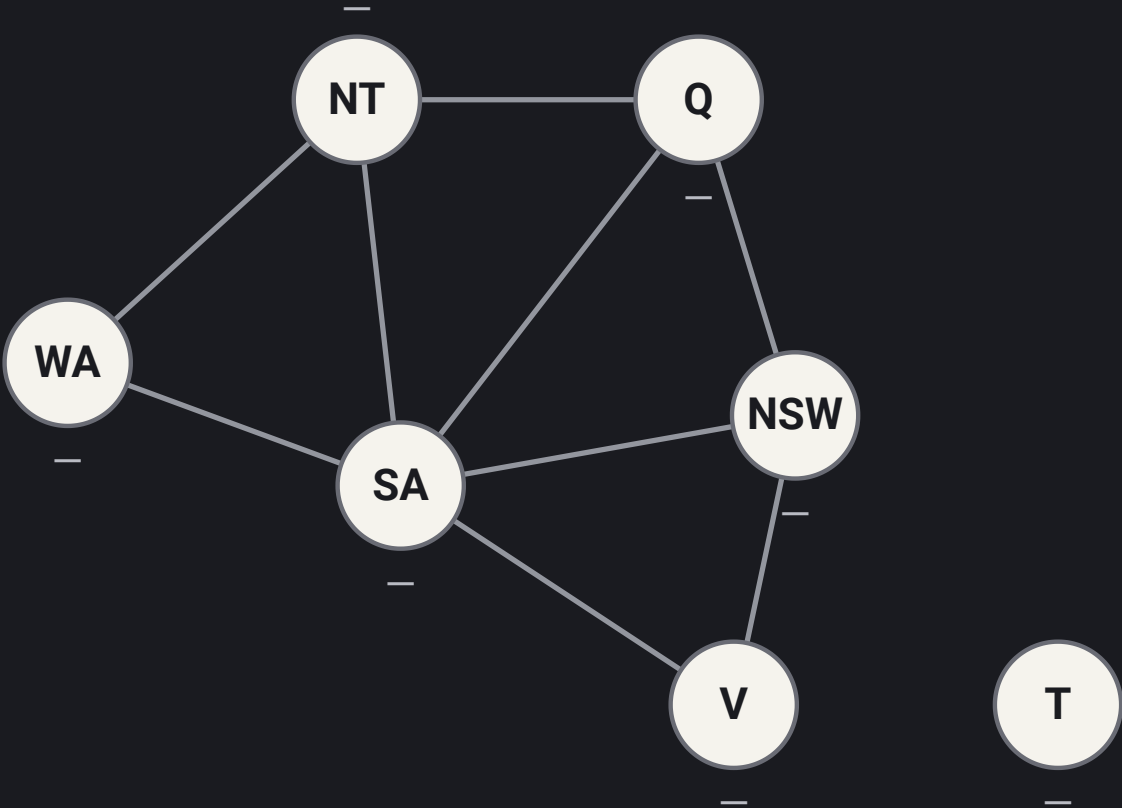
SA-Q · SA-NSW · SA-V

Q-NSW · NSW-V

T is independent

Its colour cannot create a mainland conflict.

Colour the graph and inspect the domains



Region

WA

Value

R

Assign

Clear all

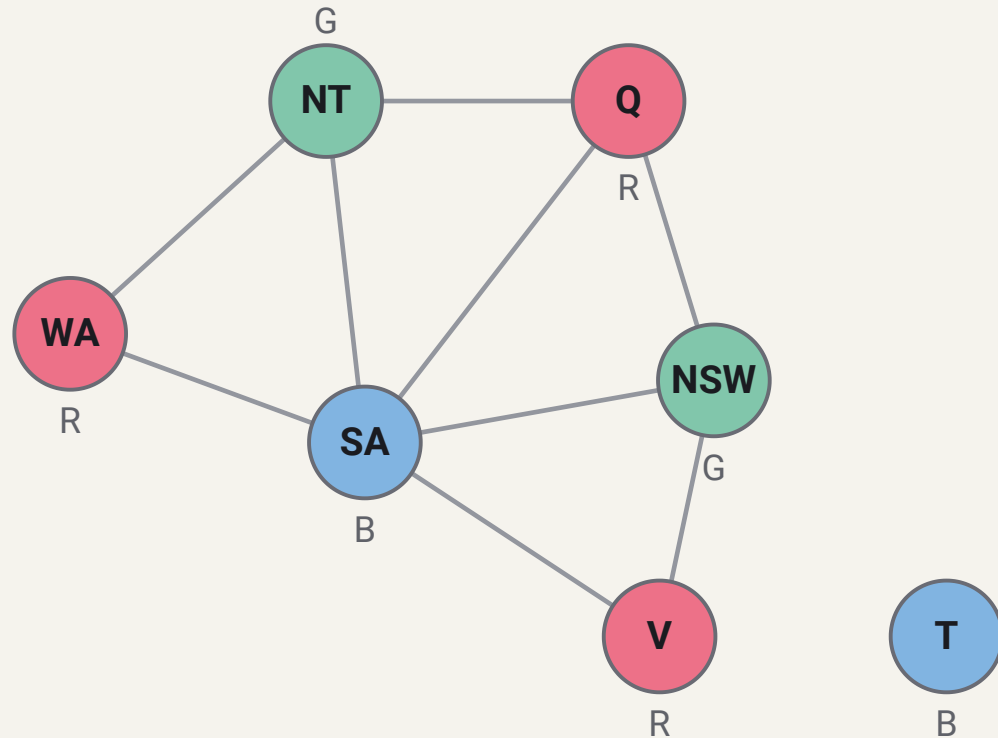
WA=R, Q=G

WA	{R, G, B}	candidates
NT	{R, G, B}	candidates
Q	{R, G, B}	candidates
SA	{R, G, B}	candidates
NSW	{R, G, B}	candidates
V	{R, G, B}	candidates
T	{R, G, B}	candidates

No assigned-edge conflict. A valid completion is not guaranteed.

Domains show filtering by assigned neighbours only. Conflicting choices remain selectable for diagnosis.

A valid colouring — and a useful failed branch



One solution

WA=R, NT=G, Q=R,
SA=B, NSW=G, V=R, T=B.

A branch to reject

WA=R and Q=G force both NT and SA to B.

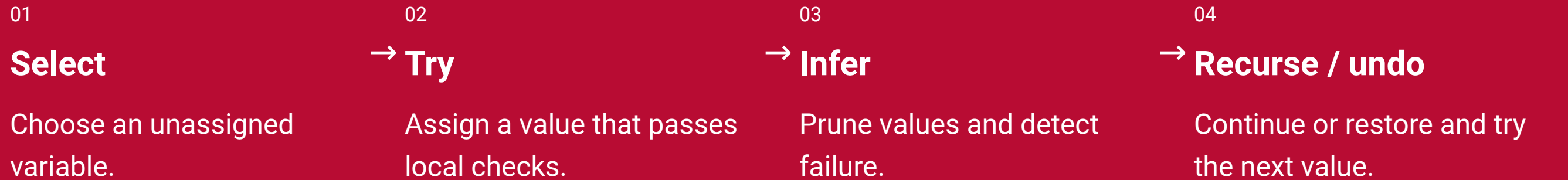
But NT and SA are adjacent.

A timetable becomes a network of decisions

Component	A deliberately small model
Variables	A, B, C: the start slot of three one-slot classes.
Domains	$A \in \{1,2\}; B \in \{1,2,3\}; C \in \{2,3\}$.
Resource conflict	$A \neq B$: they need the same lecturer.
Shared students	$B \neq C$: the same group attends both.
Precedence	$A < C$: A must finish before C begins.

Slot lengths, room capacity and overlapping durations need explicit rules in a richer model.

Backtracking builds one assignment at a time



Assign one variable per level: avoid permuting the order of the same decisions.

A solver needs an explicit undo operation

```
solve(assignment, domains):  
    if complete(assignment): return copy(assignment)  
    X ← choose_unassigned_variable(assignment, domains)  
    for value in ordered_values(X, domains):  
        if consistent(X=value, assignment):  
            mark ← trail.checkpoint()  
            assign(X, value); restrict_domain(X, {value})  
            if propagate(domains) != FAILURE:  
                result ← solve(assignment, domains)  
                if result != FAILURE: return result  
            unassign(X); trail.restore(mark)  
    return FAILURE
```

Every domain change in the failed branch must be undone before the next value.

Backtracking must restore more than the variable

Moment	D(NT)	D(SA)	Meaning
Before WA=R	{R,G,B}	{R,G,B}	Parent domains.
After forward checking	{G,B}	{G,B}	R removed in this branch.
After undoing WA=R	{R,G,B}	{R,G,B}	Restore both removals.
Trying WA=G	{R,B}	{R,B}	Filter again from the restored parent.

Keeping an old removal can turn a satisfiable instance into an incorrect “no solution”.

Watch assignment, pruning and restoration

Instance

Australia · 3 colours

Variable order

MRV + degree

Inference

Forward checking

Value order

R, G, B

Back

Next event

To result

Reset

Initial domains; no variables assigned.

Event 1/23 · Value attempts 0 · Branches undone 0

WA	{R, G, B}	candidates
NT	{R, G, B}	candidates
Q	{R, G, B}	candidates
SA	{R, G, B}	candidates
NSW	{R, G, B}	candidates
V	{R, G, B}	candidates
T	{R, G, B}	candidates

Assignment: empty

A trace event is a teaching snapshot, not a timing measurement. Ties use WA, NT, Q, SA, NSW, V, T.

What exhaustive backtracking can establish

A solution was found

The complete assignment can be checked directly against the model.

All branches were exhausted

With sound pruning and correct restoration, no solution exists.

The run was stopped

A time or node limit says only that this run did not finish.

Worst-case size

Up to d^n complete assignments for n variables with at most d values.

SAT, UNSAT and UNKNOWN are different outcomes.

MRV: expose a restricted choice early

Unassigned variable	Current legal values	Count
NT	{B}	1
SA	{B}	1
NSW	{R,B}	2
V	{R,G,B}	3
T	{R,G,B}	3

After WA=R, Q=G, MRV chooses NT or SA. A zero-value variable signals failure.

Degree breaks an MRV tie

The same snapshot

WA=R and Q=G.

NT and SA each have one candidate: B.

Unassigned neighbours

NT: SA $\rightarrow$ 1

SA: NT, NSW, V $\rightarrow$ 3

Choose SA among the MRV tie.

Use the remaining graph, not the original degree, when implementing this tie breaker.

LCV preserves options for the other variables

Try X = ...	Values removed from Y={R,G} and Z={R,B}	Score
R	Remove R from Y and R from Z.	2
G	Remove G from Y; remove nothing from Z.	1
B	Remove nothing from Y; remove B from Z.	1

For constraints $X \neq Y$ and $X \neq Z$, try G or B before R. Declare how ties are resolved.

Variable choice and value choice answer different questions

MRV + degree

Where next?

Choose a variable whose decision is likely to matter now.

LCV

Which value?

Prefer a candidate that leaves more room for future decisions.

Ordering changes the work needed to explore the tree; it does not change the model's solutions.

Forward checking looks one step ahead

On assigning $X=v$

For each unassigned neighbour Y , remove values inconsistent with $X=v$.

An empty domain rejects the branch immediately.

What it does not do

A reduced domain of Y does not automatically trigger further filtering through its other neighbours.

Forward checking reacts to the assigned variable, not to every new domain reduction.

Nonempty domains can hide an immediate conflict

After WA=R, Q=G	Forward-checking domain	Still unassigned
NT	{B}	Yes.
SA	{B}	Yes.
NSW	{R,B}	Yes.
V	{R,G,B}	Yes.

The constraint $NT \neq SA$ is impossible, although every displayed domain is nonempty.

Every value needs a supporting value

$X_i \rightarrow X_j$

For each a in $D(X_i)$, there must be some b in $D(X_j)$ satisfying the binary relation.

REVISE removes from the tail

If a has no support, delete a from $D(X_i)$.

The reverse arc must be checked separately.

Support is existential and local: different values may rely on different supports.

A directed revision, worked value by value

X candidate	Any support in $D(Y)=\{2\}$, with X	Keep?
1	$1 < 2$: yes.	Yes.
2	$2 < 2$: no.	No.
3	$3 < 2$: no.	No.

Before	After REVISE(X,Y)
$D(X)=\{1,2,3\}$, $D(Y)=\{2\}$.	$D(X)=\{1\}$, $D(Y)=\{2\}$.

AC-3 propagates changes through a queue

```
Q ← all directed arcs
while Q is not empty:
    (Xi, Xj) ← Q.pop_front()
    if REVISE(Xi, Xj):
        if D(Xi) is empty: return FAILURE
        for Xk in neighbours(Xi), Xk != Xj:
            add (Xk, Xi) to Q if not already queued
return ARC_CONSISTENT
```

A smaller $D(X_i)$ can remove supports for values in its other neighbours.

Follow the AC-3 queue

Constraint system

X < Y < Z · domains {1,2,3} ▾

Back

Revise next arc

To fixed point

Reset

All directed arcs enter the queue.

FIFO queue · next first

X → Y

Y → X

Y → Z

Z → Y

X	{1, 2, 3}	candidates
Y	{1, 2, 3}	candidates
Z	{1, 2, 3}	candidates

Exact complete assignments, independently enumerated

1 solution

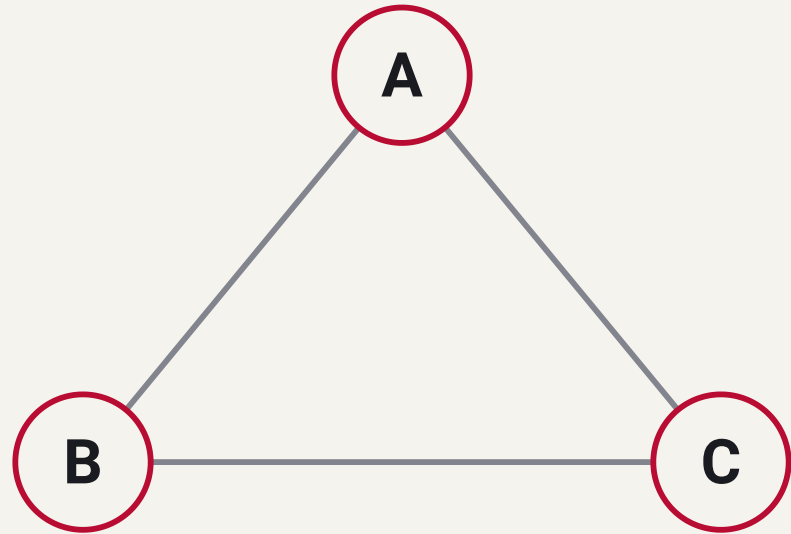
A nonempty fixed point is a filtering result. It can still have zero solutions.

A queue trace makes propagation visible

Arc revised	Deletion	Domains afterwards
$X \rightarrow Y$	X loses 3.	$X=\{1,2\}; Y=\{1,2,3\}; Z=\{1,2,3\}$
$Y \rightarrow X$	Y loses 1.	$X=\{1,2\}; Y=\{2,3\}; Z=\{1,2,3\}$
$Y \rightarrow Z$	Y loses 3.	$X=\{1,2\}; Y=\{2\}; Z=\{1,2,3\}$
$Z \rightarrow Y$	Z loses 1,2.	$X=\{1,2\}; Y=\{2\}; Z=\{3\}$
$X \rightarrow Y$	X loses 2.	$X=\{1\}; Y=\{2\}; Z=\{3\}$

The change to Y causes $X \rightarrow Y$ to be checked again. Final domains identify the unique solution.

Arc-consistent does not mean satisfiable



Edges: A–B, B–C, C–A.
Every edge requires different colours.

Only two colours: {R,G}

R always has support G on each neighbouring arc,
and G has support R.

But the triangle needs three colours

Pairwise supports cannot all be combined into one
assignment.

Empty domain $\Rightarrow$ failure. All domains nonempty after AC-3 $\nRightarrow$ solution.

MAC runs arc consistency inside backtracking

01

Decide

Choose $X=v$ and restrict its domain.

02

→ Propagate

Run AC-3 until a fixed point or empty domain.

03

→ Continue

Recurse if the branch survives.

04

→ Recover

Restore all changes after failure.

Maintaining arc consistency combines systematic search with repeated filtering.

Stronger filtering has a cost

Technique	What is checked	Practical tradeoff
Local checks	New assignment against assigned neighbours.	Cheap per decision; may discover failure deep in the tree.
Forward checking	Assigned variable against unassigned neighbours.	Prunes one step ahead.
MAC with AC-3	Supports across all affected binary arcs.	More work per decision; may save many branches.

Naive AC-3: $O(e d^3)$, where e counts directed arcs and d is the largest domain size.

AllDifferent can see a shared bottleneck

Three variables

$A \in \{1,2\}$

$B \in \{1,2\}$

$C \in \{1,2,3\}$

AllDifferent(A,B,C)

A and B consume {1,2}

The two variables must use those two values in some order.

Therefore C must be 3.

Pairwise arc consistency on $A \neq B$, $A \neq C$, $B \neq C$ does not make this deduction.

Compare binary and global propagation

Domains

A={1,2}, B={1,2}, C={1,2,3} ▾

Pairwise AC

Global GAC

Reset

Original domains. Choose a propagation method.

The global demonstration enumerates all valid triples and retains values appearing in at least one.

A	{1, 2}	candidates
B	{1, 2}	candidates
C	{1, 2, 3}	candidates

Valid distinct-value triples in the original domains

$(1, 2, 3) \cdot (2, 1, 3)$

Each button starts from the selected original domains, allowing a fair comparison.

81 decisions, 27 AllDifferent constraints

5	3			7				
6			1	9	5			
	9	8					6	

Variables and domains

One variable per cell; values 1–9.
Given digits have singleton domains.

Constraints

9 rows, 9 columns, 9 boxes.
Each group uses AllDifferent.

Illustrative givens from the source. This is not a claim of a uniquely solvable puzzle.

A candidate is not a solved cell

Cell in the illustration	Values excluded by its groups	Candidates
r1c3	Row: 3,5,7; column: 8; box: 3,5,6,8,9.	{1,2,4}
r1c4	Row: 3,5,7; column: 1; box: 1,5,7,9.	{2,4,6,8}

Candidate filtering uses the union of exclusions from the row, column and box.

One variable per column

$$Q_i \in \{1, \dots, N\}$$

Q_i is the row of the queen in column i .

Exactly one queen per column is built into the representation.

For every $i < j$

Different rows:

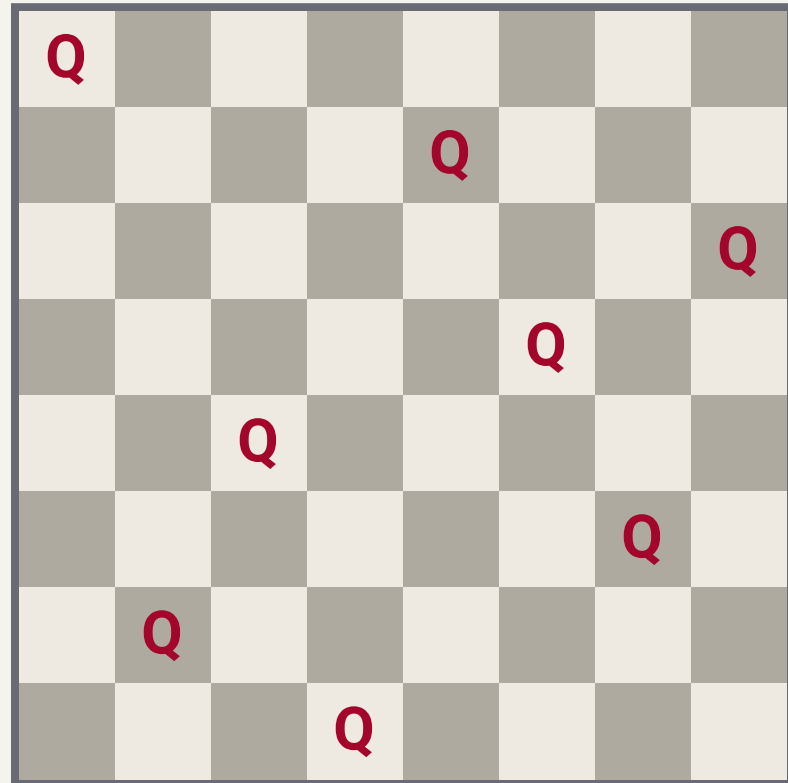
$$Q_i \neq Q_j$$

Different diagonals:

$$|Q_i - Q_j| \neq |i - j|$$

The diagonal equation compares row separation with column separation.

A solution you can validate directly



Rows by column

[1, 7, 5, 8, 2, 4, 6, 3]

Check all 28 pairs

No shared row.

No equal row and column separation.

Number of attacking pairs: **0**.

Start complete, then repair conflicts

Backtracking

Keep a partial assignment and extend it while detecting contradictions.

Local repair

Keep every variable assigned, even if constraints are violated, and change selected values.

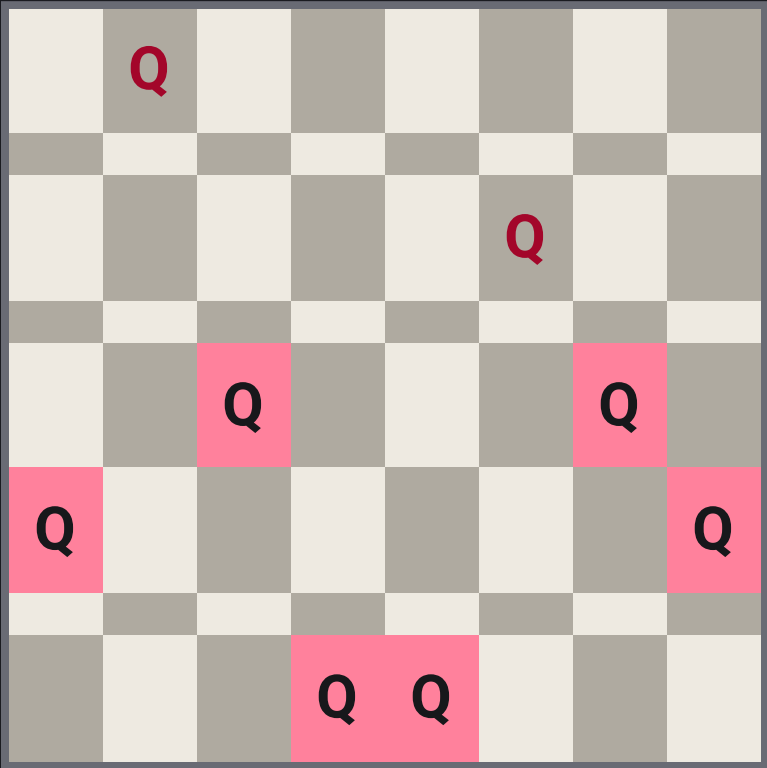
Completeness of an assignment is not completeness of an algorithm.

Min-conflicts repairs a selected variable

```
state ← a complete initial assignment
for step in 0 .. max_steps-1:
    if all constraints hold: return SOLUTION(state)
    X ← random variable involved in a violated constraint
    value ← argmin_v conflicts_if(X=v, other values fixed)
    state[X] ← value           # random tie breaking
if all constraints hold: return SOLUTION(state)
return BUDGET_EXHAUSTED
```

A best repair can leave the score unchanged. Plateaus and repeated states are possible.

Repair an eight-queen board



Click a square to move that column's queen. Rows 1–8 run from top to bottom.

Restart seed

Repair one

Run to 200 steps

Restart

Load solution

Attacking pairs / 28 possible pairs

5

Attempt 0/200. Seed 1 initialized the board.

Rows: [6, 1, 5, 8, 8, 3, 5, 6]

Read the termination status precisely

Status	What the evidence supports
SOLUTION	The current complete assignment passes every constraint.
BUDGET_EXHAUSTED	The configured number of repair attempts was used without a solution.
UNSAT after exhaustive search	Every possible branch was soundly ruled out.
Another restart succeeds	The earlier unsuccessful run was not a proof of impossibility.

Do not label a local-search timeout as “no solution exists”.

Compare methods on reproducible evidence

Hold the model fixed	Same variables, domains, constraints and stopping criteria.
Record search work	Decisions, rejected values, backtracks and domain deletions.
Measure actual cost	Wall time and peak memory, including propagation overhead.
Treat randomness honestly	Seeds, restarts, success rate and limits over multiple runs.

A short trace explains behaviour; it is not a general performance benchmark.

Disconnected components can be solved separately

Australia

The mainland and T are independent components under the land-border rule.

Combine one solution from each component.

Counting solutions

$$6 \times 3 = 18$$

Six mainland colourings and three choices for Tasmania.

For independent components, solution sets combine by a Cartesian product.

A compact model still needs all the rules

Pitfall	Correction
A missing relationship	Check every real-world conflict, precedence and capacity rule.
Domains used instead of a relation	Unary filtering is valid; cross-variable dependencies need constraints.
AllDifferent treated as a magic algorithm	State the actual propagation strength or solver configuration.
A global constraint omitted from the graph	Use a factor/hypergraph view or make its scope explicit.
Too many interchangeable representations	Add justified symmetry breaking or use a simpler variable encoding.

Small instances with known solutions and contradictions are useful model tests.

Complete the small timetable

Rule	Check for A=1, B=2, C=3
$A \in \{1,2\}, B \in \{1,2,3\}, C \in \{2,3\}$	All selected values belong to their domains.
$A \neq B$	$1 \neq 2$: satisfied.
$B \neq C$	$2 \neq 3$: satisfied.
$A < C$	$1 < 3$: satisfied.

Feasible (A,B,C) tuples: (1,3,2), (1,2,3), (2,1,3).

A preference introduces an objective

Hard constraints

Feasible assignments must satisfy resource, attendance and precedence rules.

Soft preference

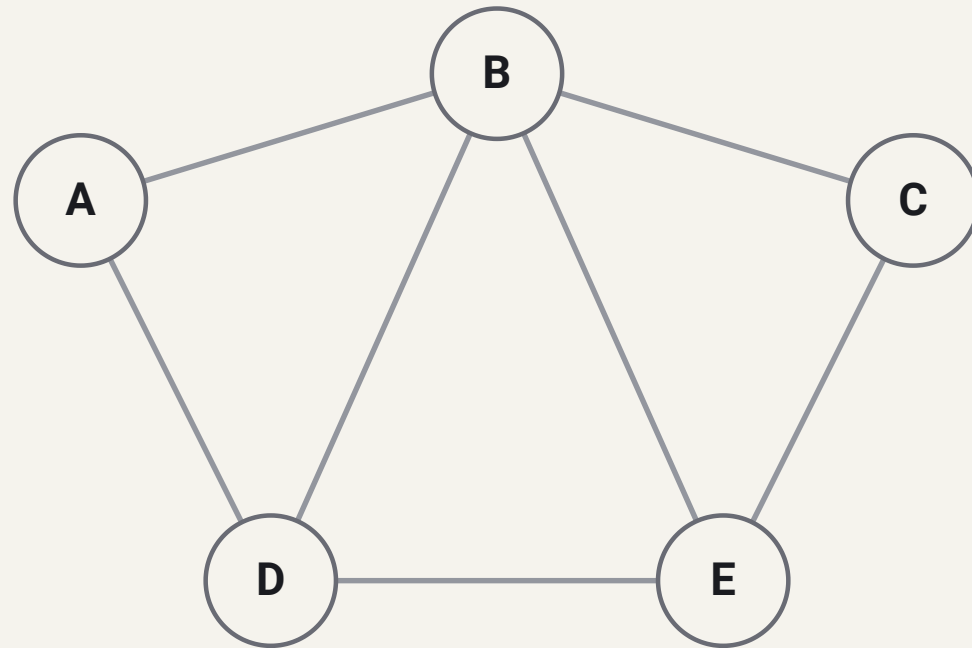
Among feasible schedules, prefer class B in an earlier slot.

minimize B

Optimal schedule: A=2, B=1, C=3.

Constraint satisfaction asks for any feasible assignment. Optimization asks for the best objective value.

Model and trace a five-region colouring



10 min

Domains: {R,G,B}.

Edges require unequal values.

1. Select the first variable by MRV + degree.
2. Try R; apply forward checking.
3. Select the next variable and finish a solution.

Edges: AB, AD, BC, BD, BE, CE, DE. Ties use A, B, C, D, E; values use R, G, B.

The first two choices expose the structure

Decision	Forward-checking effect	Next choice
B=R	A, C, D, E all become {G,B}.	MRV tie; degree tie D/E; choose D.
D=G	A and E become {B}; C stays {G,B}.	MRV tie A/E; degree chooses E.
E=B	C becomes {G}; A remains {B}.	A and C tie; alphabetic order chooses A.
A=B, C=G	All seven edges are satisfied.	Solution: A=B, B=R, C=G, D=G, E=B.

The letter B can denote a variable or the colour blue: read each assignment carefully.

Explain what propagation can and cannot prove

8 min

$X < Y$, with $D(X)=\{1,2,3\}$, $D(Y)=\{2,3\}$.

Then compare a two-colour triangle.

Revise $X \rightarrow Y$

Which X values lose all support?

Revise $Y \rightarrow X$

Does either Y value lose support now?

List solutions

Enumerate valid (X,Y) pairs.

Explain the triangle

Why can all arcs be consistent with no complete solution?

Give a supporting value for every value you keep.

Supports are local witnesses

Question	Answer
X→Y	Delete X=3. Keep 1 (supported by 2 or 3) and 2 (supported by 3).
Y→X	Keep Y=2 (supported by X=1) and Y=3 (supported by X=1 or 2).
Complete solutions	(1,2), (1,3), (2,3).
Two-colour triangle	Each edge has a different-colour support, but the three choices cannot be made simultaneously.

A support certifies one relation. A solution certifies all relations together.

Match the method to the evidence you need

Need	A reasonable starting point
Find a solution and prove absence if none exists	Complete backtracking with sound propagation and explicit limits.
Reduce branching in a finite model	MRV + degree, with FC or MAC; measure the total cost.
Exploit structured relations	Use appropriate global constraints and inspect propagation strength.
Find a feasible large assignment quickly	Consider min-conflicts when local repair fits the model; use seeds and restarts.
Optimize preferences	Add an objective and use a method that can report bounds or an optimality proof.

A good model comes first. Algorithm selection follows its structure and required guarantees.

Six questions to carry forward

01

What defines a CSP?

Variables, one domain per variable and constraints over their values.

02

Does partial consistency imply a solution?

No. A partial assignment can violate no assigned relation and still have no completion.

03

What does MRV select?

An unassigned variable with the fewest currently legal values.

04

What must backtracking restore?

The assignment and every branch-specific domain removal.

05

Does AC-3 prove satisfiability?

Not generally. A two-colour triangle is arc-consistent but unsatisfiable.

06

What does a min-conflicts limit prove?

Only that this run used its budget without a solution; it does not prove UNSAT.

Sources and further study

Supplied course material · 31 slides

01_Presentacion_Tema4_Constraint_Satisfaction_Problems(3).pptx.

Core scope, Australia, directed revision, Sudoku illustration and N-Queens.

Russell and Norvig · Artificial Intelligence: A Modern Approach

Official textbook resources · Constraint Satisfaction Problems.

UC Berkeley · CS188

Filtering and AC-3 · Ordering and structural methods

Choco-solver · Official modelling documentation

Constraints, propagators and AllDifferent alternatives

What changes when another agent opposes you?

Constraint satisfaction

Find values that satisfy a fixed collection of relationships.

choose → propagate → repair

Adversarial search

Choose actions while anticipating an opponent's responses.

minimax · alpha–beta

Next: decisions in competitive environments and how to prune a game tree safely.